Environmental fluctuation strengthens gene regulation only when it can be sensed

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Abstract

Environmental variability is widely expected to favour more elaborate gene regulation, yet models disagree even on the direction of the effect, and most measure regulatory complexity by counting network connections. We asked what fluctuation does to regulatory architecture, separating the fluctuation itself from the information an organism has with which to track it. In two independent individual-based models of genome evolution—EvoSim, whose genomes are regulatory networks of evolvable size, and R-Aevol, in which regulation emerges from sequence-encoded transcription factors—we compared frozen and fluctuating environments under an informative cue, no cue, and a yoked control that delivers the same signal molecules at the same frequencies but decoupled from the environment. In EvoSim, fluctuation raised the density of regulatory connections by 26% while lowering wiring strength per possible connection by 11%, leaving total wiring strength unchanged. In R-Aevol, a pre-registered test across three switching rates (two for the cue-dependent predictions) supported all four predictions: with an informative cue, fluctuation increased total regulatory strength 1.8–2.7-fold over a frozen environment; the effect required the cue's information rather than its mere presence (informative versus yoked, 2.3–2.9-fold); without a cue, fluctuation produced the weakest regulation of any condition; and fluctuation reduced genome size by 8–10 genes. The pattern replicated at a second population size. In EvoSim, and in R-Aevol populations with an informative cue, connection counts moved in the opposite direction to regulatory strength. [**PENDING:** one sentence on grain] Environmental fluctuation therefore does not select for regulatory complexity as such: it strengthens regulation when a cue makes the environment trackable and weakens it when none does, and counting connections can report the wrong sign.

Author summary

Organisms that live in changing environments often carry elaborate gene regulation, and it is tempting to conclude that change itself builds regulatory complexity. We tested this in two very different computer models of evolving genomes. In both, we let populations evolve in environments that were either constant or switching, and we controlled what the organisms could know about the switching: some received a signal that announced which environment they were in, some received the same signal scrambled so that it announced nothing, and some received no signal at all. We found that change alone did not build stronger regulation. When the environment switched and a meaningful signal was available, regulatory connections became much stronger; when it switched with no usable signal, they became the weakest we saw; and a

scrambled signal did not help, even though it was physically identical. We also found that simply counting regulatory connections, a common way to measure network complexity, can give the opposite answer to measuring how strong those connections are. We tested our main predictions after writing them down in advance.

Introduction

Organisms in variable environments are often thought to need, and therefore to evolve, more elaborate gene regulation. The intuition is reasonable: regulation is how a single genome produces different phenotypes in different conditions, and a constant environment offers nothing for it to do. It has also received support from comparative genomics, where the number of transcription factors scales with genome size and with the range of environments a lineage encounters [**PENDING:** citations].

Theory is precise about one level of this question and silent about another. At the level of the phenotype, whether variability favours a plastic response depends on grain and on information. A response within a lifetime is favoured when the environment changes on the timescale of an individual's life and a reliable cue predicts the state it will face [1–4]; without such a cue, fluctuation favours fixed generalist phenotypes, bet-hedging, or genetic tracking instead [3, 4]. A recent gene-regulatory-network model confirms this for the evolution of plasticity itself: plasticity evolves mostly under fast, erratic change, especially when cues are reliable, while evolvability evolves under intermediate variability and less reliable cues [5].

What this means for regulatory *architecture*—how many genes a genome carries, and how strongly they regulate one another—is much less clear, and models disagree. Tsuda and Kawata found that fluctuating selection without an environmental cue promotes complex regulatory networks, largely through the fixation of beneficial gene duplications [6] [**PENDING:** verify: their networks respond to external stimuli; confirm these carry no information about the optimum]; related models report larger genomes in fluctuating environments [7]. Work on the Aevol platform instead finds genome size bounded by selection for mutational robustness and reduced by large population size and high mutation rate [8, 9], and models of modularity find that its evolution depends on sparseness and on the structure of the variation rather than on variability as such [10, 11].

Two methodological problems stand between these results and a general answer. First, fluctuation and information are usually confounded. An environment that changes more often also offers a different amount of predictable structure, and adding a cue changes both what an organism can know and what it experiences. The standard remedy from the study of learning is the yoked control [12]: a second subject receives exactly the same stimuli as the first, but with the contingency between stimulus and outcome broken. We are not aware of it having been used in the evolution of regulatory networks. Second, regulatory complexity is usually measured by counting connections [6, 10, 11]. Counts can grow by near-neutral accumulation [13] and say nothing about how strongly a connection regulates. Weighted measures exist [14] but are rarely the primary readout.

Here we address both problems in two independent models of genome evolution that share nothing but the question. EvoSim represents a genome directly as a regulatory network whose size and connectivity evolve; R-Aevol derives regulation from transcription factors encoded in a DNA-like sequence and binding to promoter regions [15, 16]. In each we compared a frozen environment with a fluctuating one, and within fluctuation we compared an informative cue, no cue, and a yoked cue matched to the informative one in signal identity, frequency and switching rate. We measured regulatory connectivity both as a count and as a strength. After an exploratory replication in R-Aevol overturned two of our earlier conclusions from EvoSim, we stated four predictions in advance and tested them across three switching rates and two population sizes, and we pre-registered a further experiment on environmental grain to address the conflict with Tsuda and Kawata.

We find that fluctuation strengthens regulation only when a cue makes the environment trackable, weakens it when none does, and that the cue's information—not its presence—is what matters. Fine-grained fluctuation also shrinks the genome. Connection counts and connection strength can diverge in sign, so counting alone can misreport what fluctuation does to a network.

Results

Two models and one design

EvoSim is an individual-based model of single-celled organisms on a two-dimensional lattice of diffusing, cross-fed resources (Materials and methods; full ODD description in S1 Text). Each genome is a gene regulatory network—a weight matrix and node biases—of 13 to 32 nodes. Thirteen nodes form a fixed interface: twelve are read out as the phenotype (resource-uptake allocation, body size, motility, division threshold, DNA repair, stress tolerance and leakage) and one senses an abiotic cue. The remaining hidden nodes are gained by duplication and lost by deletion, so genome size evolves. The network is developed once at birth and the phenotype is fixed for life. Connectivity evolves because mutation can both create and remove connections, and each non-zero connection carries a small maintenance cost.

R-Aevol is the regulation-enabled version of the Aevol platform [15–17]. Genomes are circular sequences in which promoters, genes and binding sites are found by motif search; genes are translated into proteins that both contribute to a phenotype (a function matched against a target function) and, by binding upstream of promoters, enhance or inhibit transcription of other genes. Protein concentrations are integrated over 20 ages of an individual's life, and fitness is evaluated against the target at every age. Environmental signals are signal proteins present at a fixed concentration during the ages in which their environment is active, so an organism can evolve to read them.

The design (Fig 1) crosses environment (frozen or fluctuating) with cue (informative, none, or yoked). In R-Aevol the fluctuating environment switches between two target functions, A and B, mirror images of one another with identical geometric area, so neither is intrinsically harder. The frozen environment keeps the switching machinery but makes both targets identical. The informative arm pairs each target with its own signal protein; the yoked arm uses four environments—each target with each signal—so that the same two signals occur at the same frequencies but carry no information about the target. Because a switch in R-Aevol moves to a different environment chosen at random, a four-environment arm changes its target less often than a two-environment arm at the same switching probability; we raised the yoked arm's probability by a factor of 1.5, which equalises both the rate at which the target changes and the rate at which the signal changes (Materials and methods). EvoSim uses the same logic with an abiotic cue whose yoked version is decoupled from the environment by an incommensurate period.

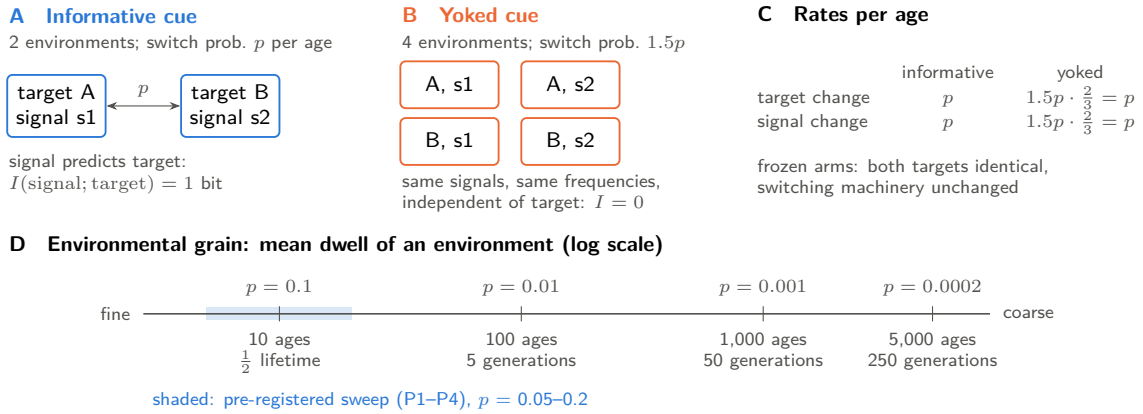


Fig 1. Design. (A) Informative arm in R-Aevol: two environments, each target paired with its own signal protein; the signal predicts the target. (B) Yoked arm: four environments pairing each target with each signal, so the same signals occur at the same frequencies but carry no information about the target. (C) Because a switch moves to a different environment chosen at random, the yoked arm switches with probability $1.5p$, which equalises both the per-age target-change and signal-change rates with the informative arm. Frozen arms use two identical targets. (D) Environmental grain as the mean dwell time of an environment, set by the switching probability p over the 20 ages of a lifetime. Ticks mark the four switching probabilities of the grain experiment (G1–G4); shading marks the range of the pre-registered sweep (P1–P4).

In EvoSim, fluctuation adds connections but not wiring strength

We first measured connectivity in EvoSim, where earlier work had shown that fluctuation raises connection density (S4 Text). In 180 populations (six arms × 30 seeds, 150,000 time steps), fluctuation without a cue raised count density—the fraction of possible connections that are non-zero—from 0.467 to 0.587 (+25.5%, $p < 10^{-4}$, $d = 2.2$; survivors only, $n = 28$ versus 30; Fig 2A). Over the same networks, strength density—summed absolute connection weight divided by the number of possible connections—fell from 0.207 to 0.184 (−10.9%, $p < 10^{-4}$, $d = -1.4$; Fig 2B). Total wiring strength did not change (38.0 versus 37.5, $p = 0.86$; Fig 2C). Fluctuation added connections without adding strength: the mean strength of a non-zero connection fell by about 30%. The same pattern held with an informative cue and with a yoked one. Node count was near the minimum of 13 in this regime and rose only slightly (+0.6 nodes, $p = 0.047$).

Two features of this regime need stating. First, fluctuating populations were 40% smaller than frozen ones, and population size can affect genome architecture on its own. The population ranges of the two arms did not overlap, so the two cannot be separated statistically; but within every arm, larger populations had *lower* strength density ($r = -0.39$; Fig 2D), so population size, taken at face value, predicts higher strength density in the smaller fluctuating populations—the opposite of what we observed. Second, in this regime the cue went unused: informative and yoked arms had identical strength density ($p = 0.99$). EvoSim organisms read the cue only when development is made reversible, and even then exploit little of its information (S4 Text), so EvoSim speaks mainly to fluctuation that cannot be tracked.

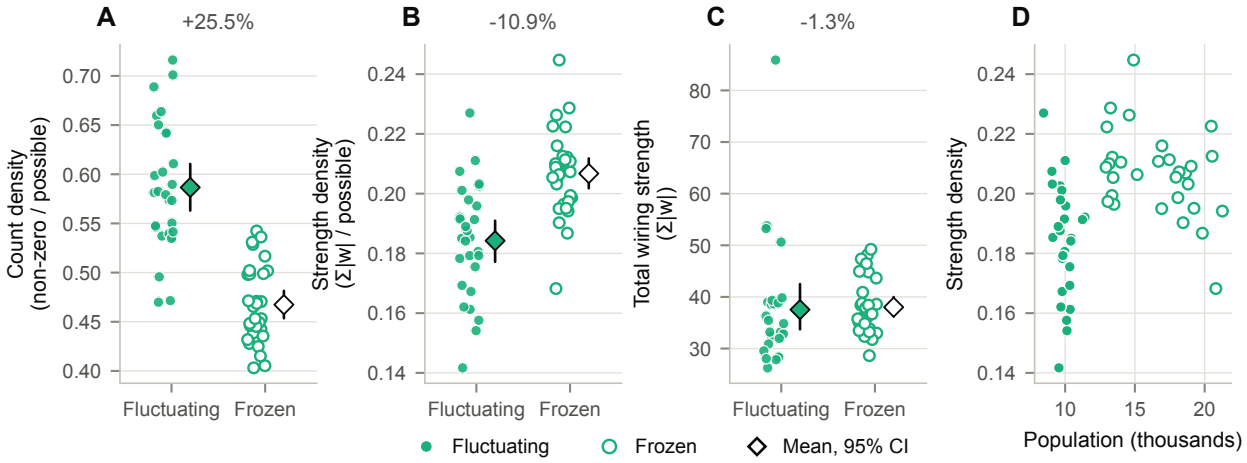


Fig 2. EvoSim: more connections, less strength. Fluctuating (no cue; filled) versus frozen (hollow) populations, one point per population; diamonds are means with bootstrap 95% confidence intervals. Numbers above panels are the change in the mean. (A) Count density: non-zero connections divided by possible connections. (B) Strength density: summed absolute weight divided by possible connections. (C) Total wiring strength. (D) Strength density against population size within each arm. Extinct populations (2 of 30 fluctuating, 0 of 30 frozen) are excluded.

In R-Aevol, whether fluctuation strengthens regulation depends on a usable cue

We next asked whether an independent model reproduced EvoSim's earlier conclusions—that fluctuation raises genome size and connectivity, and that the information content of a cue is irrelevant. It did not. In an exploratory experiment (six arms × 12 seeds, 30,000 generations), fluctuation *reduced* genome size in every cue condition, and whether it strengthened regulation depended on the cue (Fig 3). With an informative cue, fluctuating populations evolved total regulatory strength of 12.7 against 4.7 in the frozen arm with the same signals; with no cue, fluctuating populations had the weakest regulation of all six arms (1.4).

Because this interpretation was formed after seeing the data, we tested it with new seeds. Before running anything, we committed four predictions, their measures, statistical tests and decision rules to

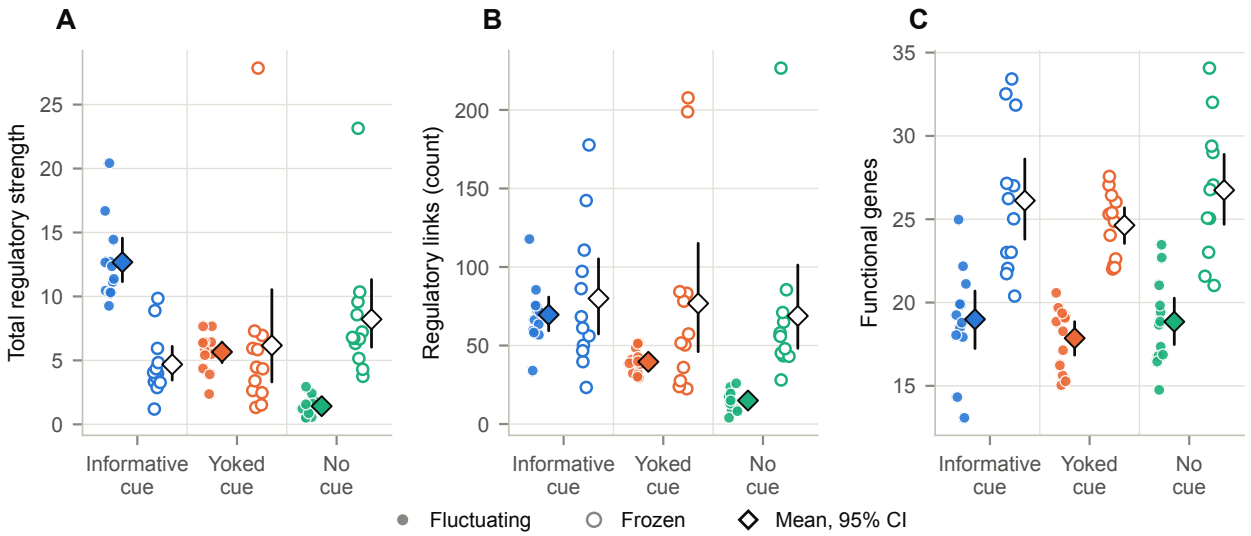


Fig 3. R-Aevol, six arms (exploratory). Total regulatory strength (A), number of regulatory links (B) and number of functional genes (C) for each combination of environment (filled, fluctuating; hollow, frozen) and cue (colour), one point per population; means with bootstrap 95% confidence intervals. Switching probability 0.10, population 1024, 12 populations per arm.

version control (Table 2; Materials and methods). The test varied the switching probability over a fourfold range (0.05, 0.10 and 0.20 per age) at a population of 1024, with 12 new seeds per cell, and repeated the central contrasts at a population of 512. For the two predictions that depend on cue use we also specified a manipulation check: that the informative cue lowers metabolic error relative to the yoked one, i.e. that organisms actually use it. The check held at switching probabilities 0.10 and 0.20 ($p < 10^{-4}$) but not at 0.05 ($p = 0.088$), so by the pre-registered rule those predictions were judged at the two faster rates only.

All four predictions were supported (Table 2, Fig 4). With an informative cue, fluctuation increased total regulatory strength over the frozen environment 1.8-fold at $p = 0.10$ and 2.7-fold at $p = 0.20$ (P1; Holm-adjusted $p = 0.005$ and $\approx 10^{-4}$). Without a cue, fluctuation reduced it roughly fivefold at every rate (1.1–1.2 against 6.3; P3; Holm-adjusted $p \approx 10^{-4}$ at each rate). The slowest-switching informative cells, excluded from the verdict, pointed the same way (11.0 against 5.9). One of these populations lost every regulatory connection for the whole analysis window, the only such case in the experiment.

The cue's information, not its presence, is what matters

The yoked control isolates information from stimulation: the informative and yoked arms receive the same two signal proteins, at the same concentration and frequency, switching at the same rate; only the contingency between signal and target differs. Informative-cue populations evolved 2.3-fold ($p = 0.10$) and 2.9-fold ($p = 0.20$) greater regulatory strength than yoked ones (P2; Holm-adjusted $p \approx 10^{-4}$ at both) and matched the target roughly twice as well (metabolic error 0.029 versus 0.058 at $p = 0.10$); in the exploratory experiment they also carried more transcription factors (11.2 versus 6.2). In a frozen environment, where the signal still switches but announces nothing because the target never changes, informative and yoked cues were indistinguishable (7.5 versus 5.4, $p = 0.18$). The difference between cue arms therefore arises only when the cue carries information about something that changes—the property the yoked design is built to isolate.

This reverses EvoSim's earlier conclusion that information content is irrelevant. The two results are compatible once cue use is taken into account: in the EvoSim regime where that conclusion was drawn, organisms exploited only about a tenth of the information available to them (S4 Text), whereas R-Aevol organisms halved their error when given an informative cue.

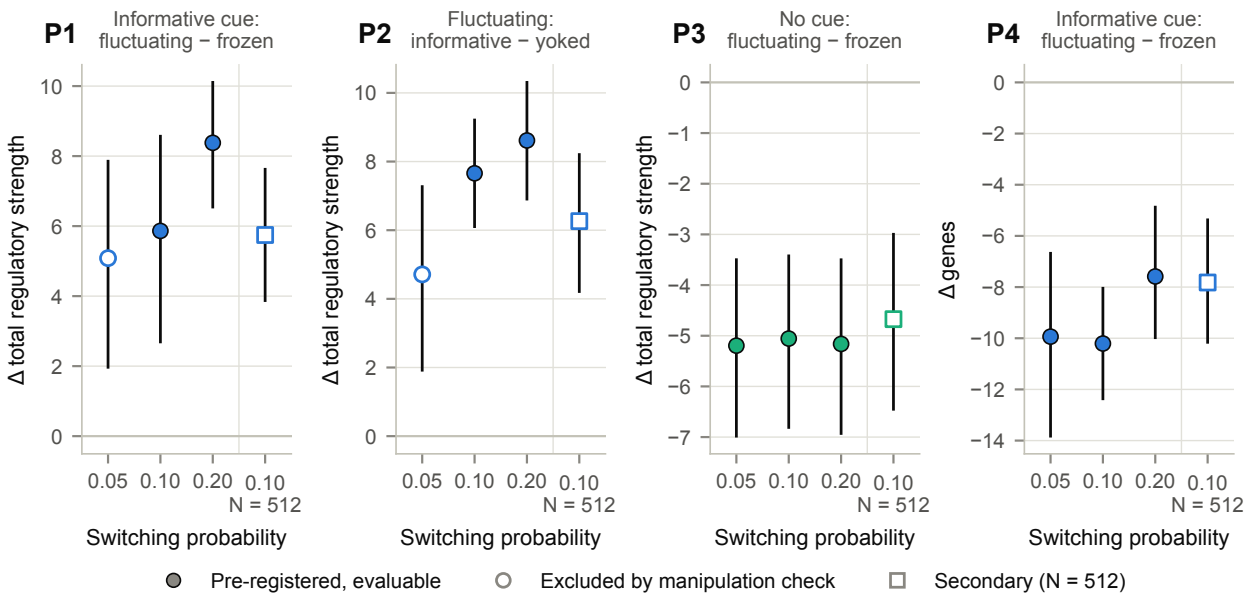


Fig 4. Pre-registered test. Differences in means with bootstrap 95% confidence intervals for each pre-registered contrast, at each switching probability (population 1024) and at population 512 (squares; secondary, no verdict). Open circles: cells excluded from the verdict because the manipulation check failed at that rate. For P3 at population 512 the frozen baseline is the population-1024 frozen arm, the only one run. P1: effect of fluctuation on total regulatory strength with an informative cue. P2: informative minus yoked cue under fluctuation. P3: effect of fluctuation without a cue. P4: effect of fluctuation on the number of functional genes.

Fine-grained fluctuation shrinks the genome

Fluctuation reduced the number of functional genes at every switching rate, by 10, 10 and 8 genes (26.3, 28.3 and 25.6 in frozen against 16.3, 18.1 and 18.0 in fluctuating populations; P4; Holm-adjusted $p \leq 3 \times 10^{-4}$; Fig 4), and at a population of 512 (25.3 against 17.5, $p < 10^{-4}$). This is the opposite of the genome enlargement reported for fluctuating selection in other models [6, 7] and of EvoSim's own earlier finding (S4 Text). One plausible mechanism is that in Aevol precision is bought with genes: the phenotype is a sum of contributions from individual genes, so a constant target that can be matched closely rewards additional genes, whereas a target that switches within a lifetime cannot be matched closely and leaves a coarser, cheaper phenotype. Frozen populations reached metabolic errors near 0.008, fluctuating ones without a usable cue near 0.06. We did not test this mechanism.

Counting connections can give the wrong sign

Connection counts and connection strength did not always move together (Fig 5). In EvoSim, fluctuation raised the count and lowered strength per possible connection. In R-Aevol, informative-cue populations in a fluctuating environment had *fewer* raw regulatory links than their frozen counterparts at every switching rate (0.60, 0.78 and 0.83 times as many), while their total regulatory strength was 1.8–2.7 times higher and their strength per link roughly twice as high. A connectivity measure based on counting links would have concluded that fluctuation with a usable cue *simplifies* regulatory networks. Without a cue, by contrast, count and strength fell together (both to about a fifth of the frozen value), so the two measures do not always disagree; the point is that a count cannot be trusted to report strength. These comparisons were specified in advance as secondary analyses and carry no verdict.

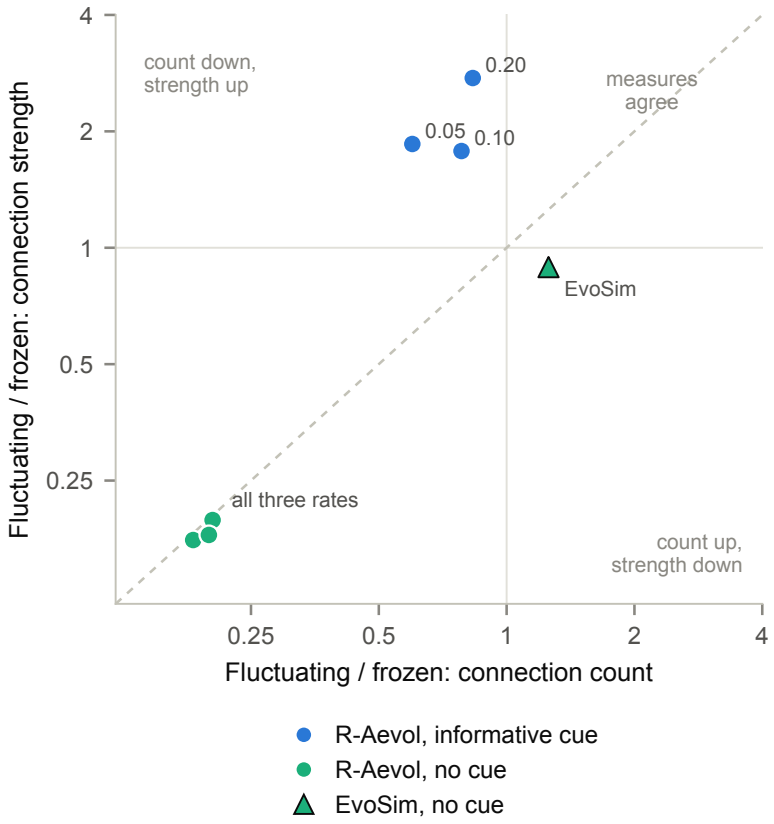


Fig 5. Counting versus weighing. Ratio of fluctuating to frozen connection count against the corresponding ratio of connection strength, on logarithmic axes. R-Aevol (circles): raw number of regulatory links and total regulatory strength, per switching probability, population 1024. EvoSim (triangle): count density and strength density, no cue. Points on the dashed diagonal are cases in which the two measures agree; points off it are cases in which counting misreports strength, and points in the upper-left or lower-right quadrants are cases in which it reports the wrong sign.

Environmental grain

To address the conflict with Tsuda and Kawata [6], who found that cue-free fluctuation across generations promotes regulatory complexity, we varied grain in R-Aevol without changing the model. The environment is a single Markov chain over ages that continues from one generation to the next, so the switching probability sets the mean time an environment persists: from half a lifetime at $p = 0.1$ to about 250 generations at $p = 0.0002$ (Fig 1D). We predicted that coarse-grained cue-free fluctuation would produce larger genomes and stronger regulation than fine-grained fluctuation (G1, G3) and than a frozen environment (G2, G4)—Tsuda and Kawata's direction—and fixed in advance how each outcome would be interpreted (Materials and methods). [**PENDING:** results of the 60-run grain experiment, expected 2026-09-28]

Discussion

Environmental fluctuation did not select for regulatory complexity as such. In R-Aevol, whether it strengthened regulation depended on whether the organism could tell which environment it was in: with an informative cue, fluctuation produced the strongest regulation we observed; with none, the weakest. A yoked cue that delivered the same molecules at the same rate did not substitute for an informative one, so it was the information, not the stimulation, that mattered. These conclusions held across switching rates

Fig 6 will show gene number and total regulatory strength against mean environmental dwell time
for cue-free fluctuating populations, with the frozen mean for reference.

Results pending (grain experiment running).

Fig 6. Environmental grain. [**PENDING:** figure and legend]

and population sizes, in a test whose predictions and analysis were fixed before the data existed.

This extends a well-established result at the level of the phenotype to the level of architecture. That reliable cues favour plastic responses and their absence favours generalism or bet-hedging is classical [1–4], and has recently been shown for the evolution of plasticity in a gene regulatory network [5]. Our contribution is to show the corresponding consequence for the network that implements the response—its regulatory strength and its size—and to separate a cue's information from its presence, which manipulations of cue reliability alone cannot do.

[**PENDING:** paragraph on grain and Tsuda & Kawata. EvoSim's fluctuation is coarse relative to a lifetime—development occurs once at birth and each environmental state lasts on the order of twenty generations—and EvoSim earlier found genome enlargement under fluctuation, as Tsuda and Kawata do; R-Aevol's fluctuation is fine-grained and shrinks the genome. The grain experiment tests whether this is the reconciling variable within one model.]

Counting connections and weighing them diverged in EvoSim and in R-Aevol's informative-cue populations, for different reasons. In EvoSim, duplication and node addition copy a node's connections with small added noise, so each copy creates many weak connections where there were none; these persist when purifying selection is weak, as it is in the smaller populations of a fluctuating environment, and connection number rises while strength does not. In R-Aevol, frozen populations carried more but weaker regulatory links, while informative-cue populations carried fewer links of roughly twice the strength; why frozen populations accumulate weak links we did not establish. In both cases the count tracked something other than the function of the network. Since connection counts are the usual measure of regulatory complexity in evolutionary models [6,10,11], we suggest that claims about connectivity report a weighted measure alongside or instead.

Limitations

- *One model for the trackable case.* The positive result—that a usable cue lets fluctuation strengthen regulation—comes from R-Aevol. EvoSim organisms read a cue only when development is reversible and exploit little of it; we have not measured regulatory strength in that regime.
- *Replication depth.* Each R-Aevol cell has 12 populations, fewer than the 60–100 common in comparable work [6,18]. The effects are large (most $d > 2$) and survive correction for multiple testing, but modest effects would be missed.
- *Measures.* R-Aevol statistics describe the best individual in each population at each generation, as the platform records them.
- *Environments.* R-Aevol's fluctuation alternates between two fixed targets. It does not include the random walk to novel optima that Tsuda and Kawata found most favourable to complexity [6].
- *Population size.* In EvoSim, fluctuation and population size are confounded across arms; the within-arm evidence argues against population as the explanation of the strength decline but cannot exclude it.
- *Genome size.* P4 confirms the direction of the genome-size effect, not its mechanism.

- *A pre-registered confirmation that could not run.* We pre-registered a confirmation of the EvoSim count–strength divergence in a regime with more hidden nodes, conditional on fluctuating populations surviving there. They did not (2 to 5 of 6 survived, against 6 of 6 frozen), so it was not run and the EvoSim result remains exploratory.
- *Timestamps.* Pre-registrations were committed to version control before the corresponding runs; for the first sweep these timestamps are local and not independently verifiable.

Conclusion

A changing world does not in itself build stronger regulation. It does so when the organism has the information to track the change, and when it has not, fluctuation leaves regulation weaker than a constant world would. Whether a model reports otherwise may depend on whether it counts connections or weighs them.

Materials and methods

EvoSim

EvoSim is described in full, following the ODD protocol [19], in S1 Text; we summarise the parts relevant here. Organisms live on a $G \times G$ lattice ($G = 128$) with six diffusing resource channels. Only the first receives external inflow; consuming channel r releases a byproduct into channel $r + 1$, with declining yields along the chain. Each organism carries an energy store, divides when it exceeds a heritable threshold, and dies when energy runs out.

Genome and development. A genome is a weight matrix W and bias vector b over n_g nodes, $13 \le n_g \le 32$. Nodes 1–12 are read out as the phenotype and node 13 receives the abiotic cue; hidden nodes $14, \ldots, n_g$ carry no fixed meaning. Development runs once, at birth: expression is initialised at 0.1 and updated eight times as

$$e_k \leftarrow \tanh\Big(b_k + \sum_m W_{mk}\, e_m + u_k\Big), \tag{1}$$

where u_k is the local concentration of resource k for $k \le 6$, the cue for $k = 13$, and zero otherwise. Expression is then mapped to the twelve traits.

Mutation. At each birth, each weight receives an additive Gaussian perturbation ($\sigma = 0.25$) with probability 10^{-3}, reduced by the organism's heritable repair trait by up to 85%. Structural mutations are gene silencing (probability 0.002; zeroes a node's outgoing weights), gene duplication (0.002; copies a node's regulatory role onto another with $\mathcal{N}(0, 0.05^2)$ noise on every copied weight), and addition or deletion of a hidden node (0.003 each; an added node duplicates an existing one in the same way). Connection silencing (probability 0.2 per birth) sets one randomly chosen weight to exactly zero; without it, additive mutation makes density ratchet towards one. Each hidden node costs 0.0015 and each non-zero weight 5×10^{-5} energy units per time step.

Environment and cue. An abiotic condition cycles through discrete states with a period of 2,000 time steps; each state rotates which resource channel pays best, and 40% of external inflow follows the favoured channel. The condition is fed to node 13. Fluctuating arms use six states, frozen arms one. The yoked cue replaces the condition's value with one generated on an incommensurate period (a multiplier of $5 + \varphi$), so that it cycles through the same values without predicting the state; the no-cue arm sets the cue's gain to zero. With a single state the cue is identically zero, so the three frozen arms are identical runs and were analysed as one.

Measures. Count density is the number of weights with $|w| > 10^{-4}$ divided by n_g^2; strength density is $\sum |w|/n_g^2$ over all n_g^2 weights, zeros included; total strength is $\sum |w|$. Each is computed per organism at birth, averaged over the population at each logged time step, and averaged over the final 20% of the run. Populations that went extinct were excluded and counted.

Runs. 180 runs: six arms × 30 seeds, 150,000 time steps each. Runs are deterministic given the seed and bit-identical across thread counts.

R-Aevol

We used the `aevol-7` branch of Aevol (commit `33277374`, https://gitlab.inria.fr/aevol/aevol) compiled with regulation enabled (`-Dwith-raevol=ON`) and per-generation statistics enabled (`-Dwithout-stats=OFF`); building on Apple Silicon required correcting a hard-coded OpenMP library path (S2 Text). Parameters (Table 1) follow the platform's own two-environment example (`examples/dynamic_env`), including its binding matrix, used unchanged in every run.

Environments. Each environment is a target function built from three Gaussians. Target B is target A reflected about the midpoint of the phenotypic axis; both have geometric area 0.152468. Fluctuating arms switch between A and B; frozen arms use two copies of the same target (A for odd seeds, B for even) so that the switching machinery runs identically but the organism experiences no change. The environment is a Markov chain over ages that continues across generations: at each of the 20 ages it moves, with probability p, to a different environment chosen uniformly among the others. We confirmed in the source that this is the switching the platform's SIMD engine performs (S2 Text).

Signals and the yoked control. Two signal proteins, taken unchanged from the platform's `complex_env` example, are present at concentration 0.9 during ages when their environment is active and absent otherwise. Informative arm: two environments, (A, s1) and (B, s2). Yoked arm: four environments, (A, s1), (A, s2), (B, s1), (B, s2). No-cue arm: no signals. In the two-environment arm every switch changes both target and signal, so both change at rate p per age. In the four-environment arm a switch changes the target with probability 2/3 and the signal with probability 2/3, so at switching probability $1.5p$ both again change at rate p per age.

Measures. R-Aevol records, each generation, statistics of the best individual. Genome size is the number of functional genes. Total regulatory strength $S = n_{\text{link}} \times \bar{a}$, where n_{link} is the number of protein–RNA influences and $\bar{a}$ their mean strength (the sum of enhancing and operating factors divided by their number); $S = 0$ when there are no links. Each run is summarised by its mean over generations 25,001–30,000.

Runs. Exploratory: six arms × 12 seeds at $p = 0.10$, population 1024 (72 runs). Pre-registered sweep: fluctuating informative, yoked and no-cue arms and frozen informative arms at $p \in \{0.05, 0.10, 0.20\}$, frozen no-cue and frozen yoked arms at $p = 0.10$ (frozen no-cue is rate-independent), 12 new seeds each, population 1024 (168 runs), plus the fluctuating informative, yoked and no-cue arms and the frozen informative arm at population 512 (48 runs). Grain experiment: cue-free fluctuating arms at $p \in \{0.1, 0.01, 0.001, 0.0002\}$ and a frozen arm, 12 new seeds each (60 runs). All completed runs reached 30,000 generations.

Statistics

The unit of analysis is a population. Each run contributes one number per measure, so generations within a run are never treated as independent. Differences between arms were tested with two-sided permutation tests on the difference of means (100,000 permutations), with rank-based permutation tests reported alongside. Multiple predictions were corrected with the Holm procedure. Effect sizes are Cohen's d; intervals in figures are bootstrap 95% confidence intervals (10,000 resamples).

Table 1. R-Aevol parameters.

Parameter	Value
Population	1024 (32×32) or 512 (32×16)
Selection	fitness-proportionate, pressure 750
Point mutation, small insertion, small deletion	5×10^{-5} per base per generation
Duplication, deletion, translocation, inversion	5×10^{-5} per base per generation
Maximum indel size	6
Initial genome	5,000 bases, one functional gene
Hill function	$n = 4$, $\theta = 0.5$
Degradation	rate 1, 10 steps per age
Lifetime	20 ages, fitness evaluated at all 20
Binding matrix	fixed file from `examples/dynamic_env`, all runs
Target sampling	300 points
Signal concentration	0.9
Generations	30,000

Pre-registration and deviations

Predictions P1–P4, their measures, tests and decision rules, and the analysis code were committed to version control (commit `f985ccd`) before the pre-registered sweep was launched; the analysis was run once, on complete data, with the committed code (verified by checksum). The only deviation was operational: the host machine's idle sleep suspended runs overnight, which lengthened wall time but cannot affect results, since runs are deterministic given their seed. The EvoSim confirmation (commit `4bc16c3`) required fluctuating and frozen populations to survive a viability pilot; none of the three candidate regimes qualified and it was not run (commit `56fe3a0`). The grain experiment was committed as `fa00777` [**PENDING:** OSF registration, if made; outcome].

Table 2. Pre-registered predictions and verdicts.

Id	Prediction	Verdict
P1	Fluctuation with an informative cue raises total regulatory strength over frozen	supported ($p = 0.10$, 0.20)
P2	An informative cue raises it over a yoked cue	supported ($p = 0.10$, 0.20)
P3	Fluctuation without a cue lowers it below frozen	supported (all rates)
P4	Fluctuation lowers the number of functional genes	supported (all rates)
P5	EvoSim: count density up and strength density down in a regime above the node floor	not run (no viable regime)
G1–G4	Coarse grain reverses the effect of cue-free fluctuation	[**PENDING:** running]

Supported: the predicted sign at every evaluable rate and Holm-adjusted $p < 0.05$ at two or more. Refuted: the opposite sign significant at any rate, or no rate Holm-significant. Otherwise inconclusive. P1 and P2 are evaluable only at rates where the informative cue measurably lowers metabolic error relative to the yoked cue.

Software and availability

EvoSim is written in Julia and depends only on the standard library for simulation. R-Aevol runs were made on a single 12-core machine. All code—both models' configurations, launchers, pre-registered analysis scripts and the full history of the analyses—and the per-run statistics are available at [**PENDING:** repository URL and Zenodo DOI].

Supporting information

S1 Text. ODD description of EvoSim. [**PENDING:** to be written]

S2 Text. R-Aevol build, configuration files and the yoked-control derivation. [**PENDING:** to be written]

S3 Table. Per-arm and per-run results for every experiment. [**PENDING:** to be assembled]

S4 Text. Earlier EvoSim results and the audit trail. Claims that were superseded, the measurement traps that produced them, and the control that caught each. [**PENDING:** to be written]

S5 Table. Prior-art comparison. [**PENDING:** to be assembled]

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[**PENDING:** acknowledgements; disclosure of AI-assisted analysis and writing, as required by PLOS policy]

References

1. Levins R. Evolution in Changing Environments. Princeton: Princeton University Press; 1968.
2. Moran NA. The evolutionary maintenance of alternative phenotypes. Am Nat. 1992;139(5):971-89.
3. Kussell E, Leibler S. Phenotypic diversity, population growth, and information in fluctuating environments. Science. 2005;309(5743):2075-8.
4. Botero CA, Weissing FJ, Wright J, Rubenstein DR. Evolutionary tipping points in the capacity to adapt to environmental change. Proc Natl Acad Sci USA. 2015;112(1):184-9.
5. Jorritsma A, van den Berg P. The evolution of plasticity and evolvability in a simple gene regulatory network. Evolution. 2026;80(1):230-9. doi:10.1093/evolut/qpaf219.
6. Tsuda ME, Kawata M. Evolution of gene regulatory networks by fluctuating selection and intrinsic constraints. PLoS Comput Biol. 2010;6(8):e1000873.
7. Bentkowski P, van Oosterhout C, Mock T. A model of genome size evolution for prokaryotes in stable and fluctuating environments. Genome Biol Evol. 2015;7(8):2344-51.
8. Knibbe C, Coulon A, Mazet O, Fayard JM, Beslon G. A long-term evolutionary pressure on the amount of noncoding DNA. Mol Biol Evol. 2007;24(10):2344-53. [verify].
9. Luiselli J, Rouzaud-Cornabas J, Lartillot N, Beslon G. Genome streamlining: effect of mutation rate and population size on genome size reduction. Genome Biol Evol. 2024;16(12):evae250.
10. Espinosa-Soto C. On the role of sparseness in the evolution of modularity in gene regulatory networks. PLoS Comput Biol. 2018;14(5):e1006172.
11. Kashtan N, Alon U. Spontaneous evolution of modularity and network motifs. Proc Natl Acad Sci USA. 2005;102(39):13773-8.
12. Church RM. Systematic effect of random error in the yoked control design. Psychol Bull. 1964;62:122-31.

13. Soyer OS, Bonhoeffer S. Evolution of complexity in signaling pathways. Proc Natl Acad Sci USA. 2006;103(44):16337-42.

14. Iwasaki WM, Tsuda ME, Kawata M. Genetic and environmental factors affecting cryptic variations in gene regulatory networks. BMC Evol Biol. 2013;13:91.

15. Sanchez-Dehesa Y, Parsons D, Peña JM, Beslon G. Modelling evolution of regulatory networks in artificial bacteria. Math Model Nat Phenom. 2008;3(2):27-66.

16. Beslon G, Sanchez-Dehesa Y, Parsons DP, Peña JM, Knibbe C. Scaling laws in bacterial genomes: a side-effect of selection of mutational robustness? BioSystems. 2010;102(1):32-40. [verify].

17. Luiselli J, Parsons DP, Gallé R, Banse P, Rouzaud-Cornabas J, Beslon G. Aevol-9: a simulation platform to decipher the evolution of genome architecture. bioRxiv [Preprint]. 2025. doi:10.1101/2025.04.10.648095.

18. Lalejini A, Ferguson AJ, Grant NA, Ofria C. Adaptive phenotypic plasticity stabilizes evolution in fluctuating environments. Front Ecol Evol. 2021;9:715381.

19. Grimm V, Railsback SF, Vincenot CE, Berger U, Gallagher C, DeAngelis DL, et al. The ODD protocol for describing agent-based and other simulation models: a second update to improve clarity, replication, and structural realism. J Artif Soc Soc Simul. 2020;23(2):7.